

Mobina Noorani

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RESEARCH INTERESTS

My research focuses on building intelligent, reasoning-capable AI systems that can perceive, retrieve, and act across multiple modalities. I am particularly interested in reinforcement learning for large foundation models, agentic AI systems with tool-use and self-verification, multimodal and vision-language models, knowledge-grounded reasoning, generative AI, RAG, and computer vision — with the goal of developing AI that can reliably reason and generalize in complex real-world environments.

EDUCATION

M.Sc — Electrical & Computer Engineering — University of Florida 2023 – Present

Research: Multimodal Learning, Computer Vision, Agentic AI, LLMs, Clustering Algorithms, Reinforcement Learning
Coursework: Computer Vision, Pattern Recognition, Machine Learning, Image Processing, Urban Analysis, Computer Communication

M.Sc — Artificial Intelligence — University of Iran | GPA: 4.0 / 4.0 2018 – 2021

Research: Computer Vision, Image Processing, Neural Network Optimization

B.Sc — Information Technology — University of Semnan | GPA: 4.0 / 4.0 2013 – 2017

Research: Web Development, Network Engineering

RESEARCH EXPERIENCE

Graduate Research Assistant | University of Florida Gainesville, FL | Sep 2023 – Present

- Currently developing an agentic AI pipeline to detect and classify key architectural elements (roof condition, sill height, building typology) from large-scale satellite and street-view imagery using vision-language models and goal-directed automation
- Training a large vision-language model using reinforcement learning (GRPO) to autonomously reason, retrieve external knowledge, and self-verify answers for knowledge-intensive visual question answering — deployed on multi-GPU HPC infrastructure using FSDP, vLLM, and Ray
- Developed RepSC-SOM, a novel spatially-constrained self-organizing map extending standard SOM with representative initialization, adaptive geographic filtering, and region-growing refinement — outperforming standard baselines on real-world contamination hotspot detection
- Developed an agentic AI planning support system built on RepSC-SOM, integrating LLM-based reasoning and human-in-the-loop interaction to guide demand-oriented spatial regionalization for disaster adaptation planning

Teaching Assistant | University of Florida Gainesville, FL | May 2024 – Aug 2025

- Provided technical support for coursework including Python programming, data preprocessing, and applying machine learning and deep learning models to different datasets

Teaching & Research Assistant | Alfred University Alfred, NY | Jan 2023 – Aug 2023

- Contributed to quantum ML research, assisting in the development and testing of quantum algorithms for ML applications
- Supported Computer Programming course: graded assignments, mentored 30+ students on programming fundamentals

Teaching Assistant | University of Iran Tehran, Iran | Oct 2021 – Mar 2022

- Instructed Computer Vision course covering image processing, feature extraction, and deep learning methods; guided 20+ students through final projects implementing CNNs and object detection models

PUBLICATIONS & PRESENTATIONS

- **Demand-Oriented Regionalization with Local Data** *Mobina Noorani, Shangde Gao, Changjie Chen, Karla Saldana Ochoa* Scientific Reports (Nature Portfolio) (Published 2026) <https://www.nature.com/articles/s41598-026-47764-4>
- **Promising Applications for Promoting Disaster Resilience on College Campuses** *Mobina Noorani, Changjie Chen, Kaleb Smith, Fernando Fernandez, Karla Saldana Ochoa* The Digitized Campus: AI and Big Data in Higher Education, SUNY Press (Published 2026) <https://sunypress.edu/Books/T/The-Digitized-Campus>
- **From Shading Masks to Surrogate Frameworks: A Topological Approach for Performance-Driven Envelope Design** *Julia Gomez Goenaga, Shangde Gao, Aurora Monge-Barrio, Seyedeh Mobina Noorani, Antonio Villanueva Peñalver, Bassam Nory, Karla Saldana Ochoa* Available at SSRN (preprint) (2025) <https://ssrn.com/abstract=6530551>
- **Enhancing Demand-Oriented Regionalization with Agentic AI and Local Heterogeneous Data for Adaptation Planning** *Mobina Noorani, Shangde Gao, Changjie Chen, Karla Saldana Ochoa* NeurIPS 2025 Workshop — UrbanAI: Harnessing Artificial Intelligence for Smart Cities (peer-reviewed) (Published 2025) <https://openreview.net/forum?id=bKfHuyel9L>
- **Geo-SOM and Hierarchical Clustering for Flood Risk Assessment** *Mobina Noorani et al.* Optimizing Solutions for Resilient Coasts Summit (Poster) (Dec 2024) <https://ccs.eng.ufl.edu/optimizing-solutions-for-resilient-coasts-summit/>
- **Campus Flood Assessment through a Web-Interface Powered by AI** *Mobina Noorani, Changjie Chen, Kaleb Smith, Karla Saldana Ochoa* Symposium on Flooding Adaptation (Poster) (Oct 2024) <https://conference.ifas.ufl.edu/flooding/documents/posters/3-Mobina-Noorani-Poster.pdf>

KEY PROJECTS

Multimodal VQA with Reinforcement Learning & Agentic Retrieval

2025 – Present

- Training a large vision-language model using GRPO to autonomously reason over images, decide when to retrieve external knowledge, and self-verify answers for knowledge-intensive VQA tasks
- Designed a multi-turn agentic pipeline integrating dense retrieval (FAISS over 21M Wikipedia passages) with RL-trained reasoning — model learns when and how to search through outcome-only reward
- Deployed on 4× NVIDIA B200 GPUs using FSDP, vLLM, and Ray on UF HiPerGator; tracked with W&B — ongoing research targeting a peer-reviewed publication

Agentic AI Planning Support System

2025

- Created an agentic AI framework integrating LLM-based reasoning, planning, and human-in-the-loop interaction to generate demand-oriented regionalization for disaster adaptation planning
- Built on top of RepSC-SOM, enabling planners to interactively refine spatial regions through natural language guidance and AI-assisted constraint exploration

Digital Twin Project: RepSC-SOM

2024 – 2025

- Developed RepSC-SOM, a SOM-based regionalization model with representative initialization, adaptive geographic filtering, and region-growing refinement
- Applied to flood-induced E. coli contamination data in Jacksonville, demonstrating more accurate detection of contamination hotspots compared to census tracts and standard units

Campus Vulnerability — Object Detection for Flood Risk

2024 – Present

- Developing an object detection system to identify and classify flood-prone architectural features from imagery for campus vulnerability assessment

Facial Trait Recognition, Emotion Detection & Sentiment Analysis

2021

- Designed and trained deep learning models to predict 42 psychological attributes from facial images, introducing a novel activation function that improved emotion recognition performance
- Developed a multimodal sentiment analysis system combining facial image analysis with text data from Twitter, demonstrating early research experience in cross-modal AI

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS

ML / DL Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV

AI Systems: Agentic AI, RAG, Multimodal Learning, LLM Fine-tuning, RL / GRPO / Reward Modeling, VLMs

Infrastructure: Multi-GPU HPC (SLURM), CUDA, vLLM, FlashAttention, FSDP, Ray, FAISS, Distributed Training
Tools: Git, Jupyter, W&B

INDUSTRY EXPERIENCE

Web & Backend Developer | Pardis Tehran Science & Technology ParkTehran, Iran | Nov 2018 – Dec 2019

- Designed and shipped full-stack web applications (PHP, JavaScript, MySQL) in a startup environment; built REST APIs and data-driven backend systems with Git-based team workflows

CERTIFICATIONS & COURSES

Agentic AI — DeepLearning.AI (Andrew Ng)	Nov 2025
Retrieval-Augmented Generation (RAG) — DeepLearning.AI	Dec 2025
Fine-tuning & RL for LLMs: Intro to Post-training — Sharon Zhou, DeepLearning.AI	In Progress
Generative Adversarial Networks Specialization — Sharon Zhou, Coursera	Completed
Generative AI with Large Language Models — Chris Fregly, Coursera	Completed
Quantum Machine Learning — University of Toronto / edX	Aug 2023
Custom Models, Layers & Loss Functions with TensorFlow — Laurence Moroney, Coursera	Apr 2022
Convolutional Neural Networks — Andrew Ng, Coursera	Apr 2021
Structuring Machine Learning Projects — Andrew Ng, Coursera	Aug 2020
Improving Deep Neural Networks — Andrew Ng, Coursera	Jul 2020
Neural Networks and Deep Learning — Andrew Ng, Coursera	May 2020

REFERENCES

Dr. Hassan Azad — University of Florida — h.azad@ufl.edu
Dr. Damon Woodard — University of Florida — dwoodard@ece.ufl.edu